UNIVERSITY OF BAGHDAD

COLLEGE OF MEDICINE

2026-2027

Introduction to nutrition

Department of Family and Community Medicine

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Learning Objectives

- Understand the relevant definitions.
- Define the staple food
- List the main nutrient in food and classify them accordingly.
- Define energy density, nutrient density and how to calculate them.

  Explain dietary guidelines.

  Distinguish between 4 set of dietary reference intake.

Definitions:

❖**Nutrition:** The science of **food and nutrients,** their ingestion, digestion, absorption, transport, metabolism, interaction, storage, and excretion for meeting energy requirement, building, growing, maintaining tissues to promote optimum health.

❖ **Clinical nutrition:** Is a medical specialty dealing with the relationship between disease & nutrition.

Acute & chronic illness is caused by deficiencies or excesses of dietary components.

❖**Nutrients: Substances obtained from food to meat the body need** (i.e.; building, growing, maintaining tissues…)

Staple food: A food that forms the **main basis of the diet**, consumed **frequently and in relatively large amounts**, and provides **a substantial proportion of dietary energy**.

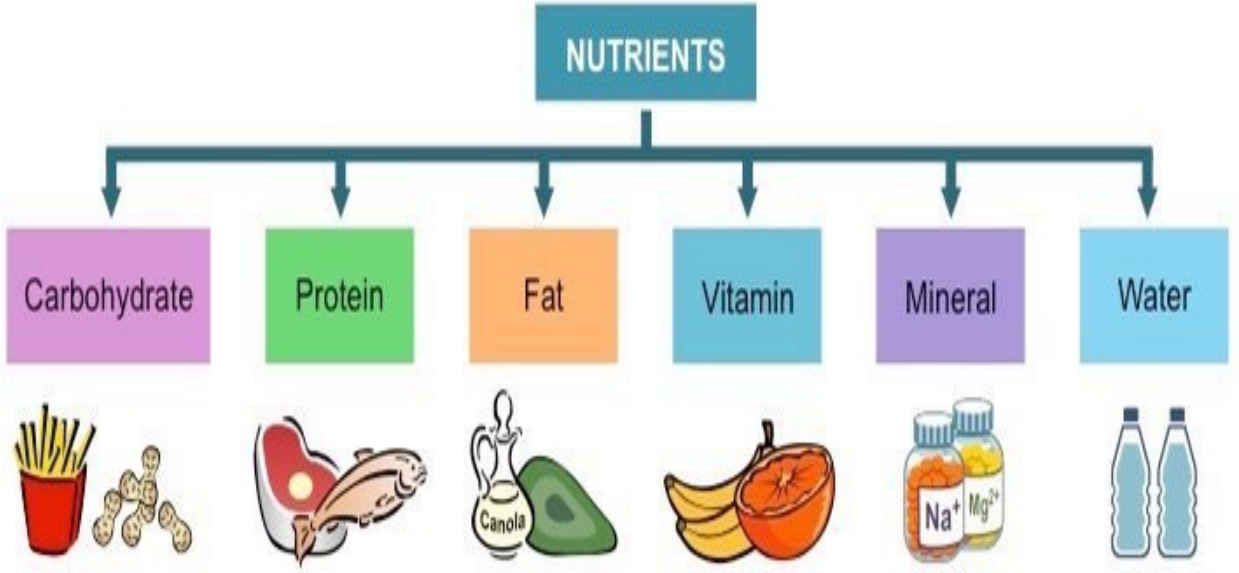
Staple foods vary by **geographical region** and commonly include **rice, wheat, maize, and other cereals**.





Six Classes of Nutrients:

- ✓Carbohydrates
- ✓Proteins
- ✓Fats
- ✓Vitamins
- ✓Minerals
- ✓water



There are several ways to classify the classes of nutrients:

1. Organic or inorganic
2. Essential or nonessential
3. Macronutrient or micronutrient
4. Energy yielding or not

1. Classifying nutrients by their chemical composition:

- **Organic nutrients:** in chemistry, substances or molecules containing **carbon-carbon bonds (C-C) or carbon-hydrogen bonds(C-H).** The **four** organic nutrients are **carbohydrate, fat, protein, and vitamins.**
- **Inorganic nutrients: not containing carbon**. The **two** classes of nutrients that are inorganic are **minerals and water.**

2. Classifying nutrients by their essentiality:

- **Essential nutrients:** nutrients a person **must** obtain from food because the body cannot make them for itself in **sufficient quantity** to meet physiologic needs.

Biotin, Vitamin D, Vitamin K2

- **Nonessential nutrient:** nutrient that **can be** manufactured in the body by means of other nutrients. Thus, it is not essential to consume this nutrient regularly in the diet. for example: **cholesterol, and some a.a (alanin, asparagine, aspartic acid, glutamine….)**

3. Classifying nutrients by their quantity needed:

- **Macronutrient:** are nutrients the body needs in **large amounts** to function optimally include **carbohydrates, proteins, fats, and water.**

- **Micronutrients:** are nutrients the body needs in relatively **small amounts** include **vitamins and minerals.**

4. Classifying nutrients by energy-yielding:

- **The three** energy-yielding nutrients are **carbohydrate, protein, and fat.**

- **Vitamins and Minerals** do **NOT** provide energy to the body. But they facilitate the release of energy from the three energy-yielding nutrients.

- **Water** also does **NOT** provide energy but it is the medium in which all of the body's processes take place.

- **1 kcalorie (kcal):** is the amount of heat necessary

to raise the temperature of 1 kilogram (kg) of water 1°C.

- **1 kcalorie = 1 Calorie = 1000 calories**
- **Calories in nutrition:** refer to the amount of energy provided by food or expended by the body during physical activity.

- ✓Each **1 gram of carbohydrate** yields **4 kcalories .**
- ✓Each **1 gram of protein** yields **4 kcalories**.
- ✓Each **1 gram of fat** yields **9 kcalories.**
- ✓Each **1 gram of alcohol** yields **7 kcalories.**

(Alcohol is **not a nutrient** because it cannot support the body's growth, maintenance, or repair).

Energy density vs Nutrient density:

Energy density: a measure of the energy a food provides relative to the amount of food (kcal per gram).

Energy density = calories ÷ weight

Lower energy density foods provide less energy per gram of food – this means that you can have larger, more satisfying portions of these foods for fewer calories.

- **Nutrient density:** refers to the amount of **nutrients (vitamins, minerals, fiber, high-quality proteins, and healthy fat …)** provided by a food relative to the amount of energy(calories) it provides.

$$\text{Nutrient Density} = \frac{\text{Nutrient Content (vitamins, minerals, protein, etc.)}}{\text{Energy Content (Calories or kcal)}}$$

- **Nutrient dense foods or drinks:** include foods and drinks that provide a **high amount of nutrients relative to its calories content** for example; vegetables, fruits, legumes, whole grains, lean meats, fish, skinless poultry, and low-fat or fat-free dairy products.

❑ Energy-Dense foods or drinks:

- ✓ **Energy dense, nutrient poor foods or drinks**: include foods and drinks that provide **little or no nutrient relative to their calorie content**, These are often referred to as **"empty-calorie"** e.g.; sugary drinks, candies, carbohydrate-based desserts, doughnuts, pastries, and some chocolate-based desserts.
- ✓ **Energy dense, nutrient dense foods and drinks:** include foods and drinks that provide a lot of both **calories and nutrients** such as avocados, cheese, seeds (e.g., chia seeds, sunflower seeds), nuts, nut butters

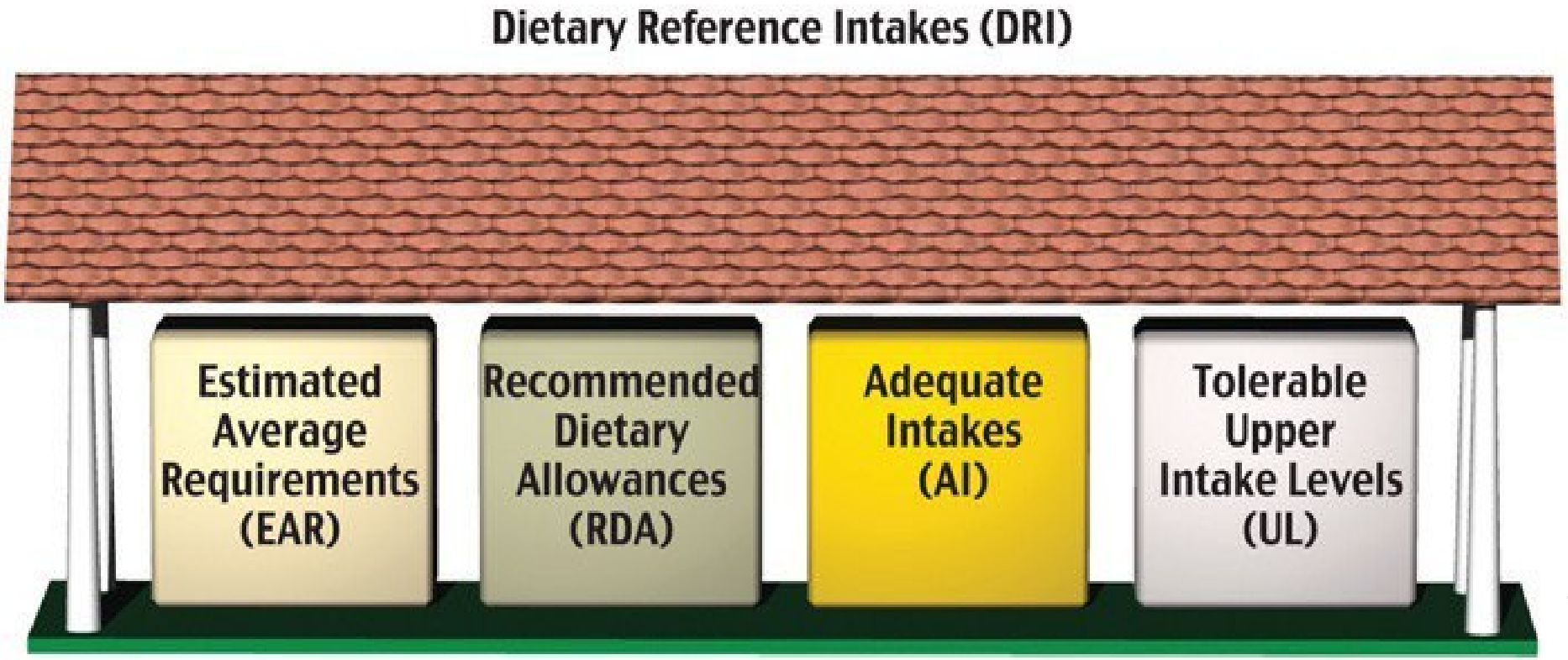
Q: A100-gram serving of plain, low-fat Greek yogurt contains 115 mg of calcium and 73 calories?

Q: Calculating and Comparing Energy Density of two different snacks, both weighing 150 grams:
Snack A (Glazed Doughnut): Contains 420 calories.
Snack B (Fresh Apple): Contains 78 calories?

Q: An athlete wants to maximize protein intake relative to total calorie consumption. She compares two options:
Option 1(Cottage Cheese - 110g): Contains 90 calories and 12 grams of protein.
Option 2 (Whole Milk - 240g): Contains 150 calories and 8 grams of protein.
Calculate the protein density (g of protein per 100 kcal) for both items to determine which is more nutrient-dense?

Dietary reference intake(DRI):

A set of 4 types of reference values used to assess and plan nutrient intakes for healthy individuals **of specific age and gender group..**



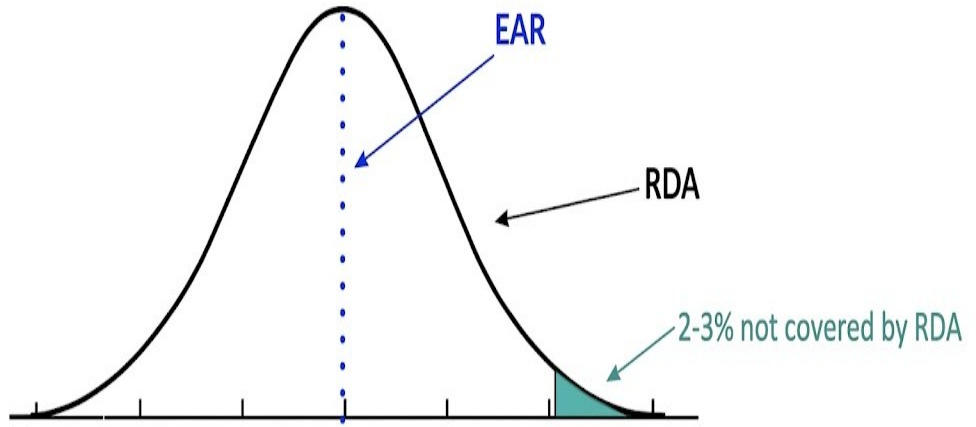
(1)Estimated Average Requirements (EAR): This is the intake level that meets the needs of **half (50%) of healthy individuals in a specific age and sex group**;

- ✓ It is used to assess the adequacy of nutrient intakes in **populations,** and
- ✓ Serves as the basis for establishing RDA valu



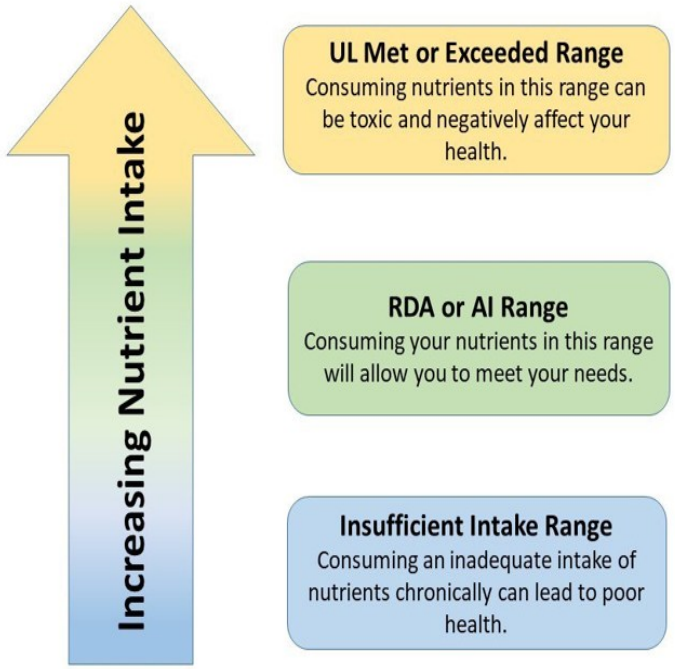
(2)Recommended Dietary Allowances (RDA): presents the average daily nutrients needed to meet the requirements of almost all **(97%- 98%)** of the healthy individuals for a given age and gender.

- An RDA for a nutrient should serve as a goal for intake for **individuals,** not as a benchmark for adequacy of diets of populations.



(3)Adequate Intakes (AI): The Adequate Intake is used as a guide when insufficient scientific evidence is available to establish the RDA. Both the RDA and the Adequate Intake may be used as goals for individual intake.

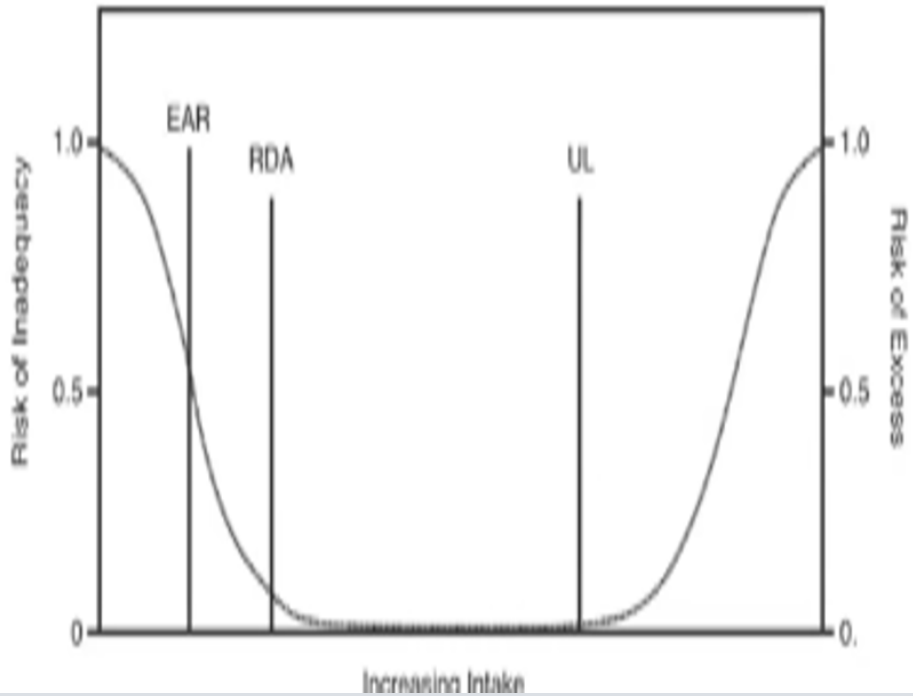
(4)Tolerable Upper Intake Levels (UL): This indicator is not a recommended intake. Rather, it sets the maximal intake that is unlikely to pose adverse health risks in **almost all** healthy individuals **in an age- sex specific group.** As intake increases above the UL, the potential risk of adverse health effects increases.



UL Met or Exceeded Range
Consuming nutrients in this range can be toxic and negatively affect your health.

RDA or AI Range
Consuming your nutrients in this range will allow you to meet your needs.

Insufficient Intake Range
Consuming an inadequate intake of nutrients chronically can lead to poor health.



Levels of Nutritional Status :

(1)Ideal nutrition :

- Nutrient intake is **sufficient to meet daily requirements** and **maintain adequate body stores**, while maintaining a **healthy body weight**.

(2)Borderline nutrition:

- Nutrient intake is **sufficient to meet daily requirements**, but **insufficient to maintain adequate nutrient reserves/ store**.
- Nutrient stores may be inadequate to meet the **increased demands associated with stress**, such as **illness, pregnancy, growth, and development**.

(3) Malnutrition:

A. Undernutrition_Two Aspects:

1. Inadequate energy intake:

- **Energy** intake is **insufficient to meet the body's requirements**, resulting in **weight loss and depletion of body energy stores**.
- Severe undernutrition can be **life-threatening** and may require **medical intervention**.

2. Specific nutrient deficiency:

- Results from a **relative or absolute deficiency of a specific nutrient**.
- Examples: **vitamin A deficiency** and **iron-deficiency anemia, protein def.**

Overnutrition – Two Aspects:

1. Excess energy intake:

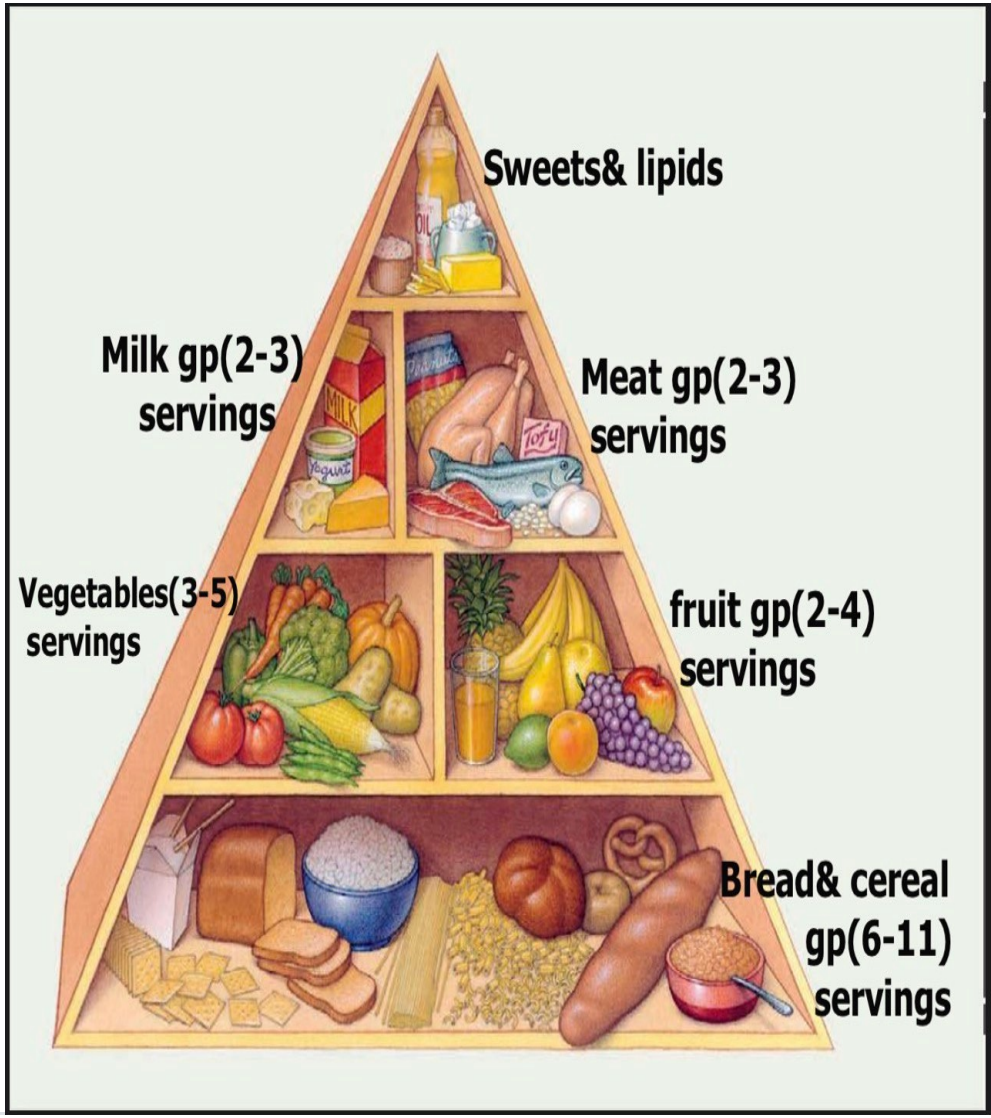
Excessive **energy** intake over time can lead to **overweight and obesity**, increasing the risk of obesity-related diseases such as **cardiovascular disease and type 2 diabetes**.

2. Excessive intake of specific nutrients:

Excessive intake of **certain nutrients**, particularly through **high-dose vitamin or mineral supplements**, can lead to **nutrient toxicity**.

Dietary guidelines: (according to food groups)

A food pyramid is a representation of the **optimal number** of **servings** to be eaten each day from each of the basic **food groups.** In 1992 pyramid introduced by the **United States Department of Agriculture (USDA)** was called the **"Food Guide Pyramid" or "Eating Right Pyramid".**



Composition of USDA pyramid:

✓**6-11 servings/day** of bread, cereal, rice and pasta occupied the **large base** of the pyramid

✓Followed by **3 to 5** servings of **vegetables/day**

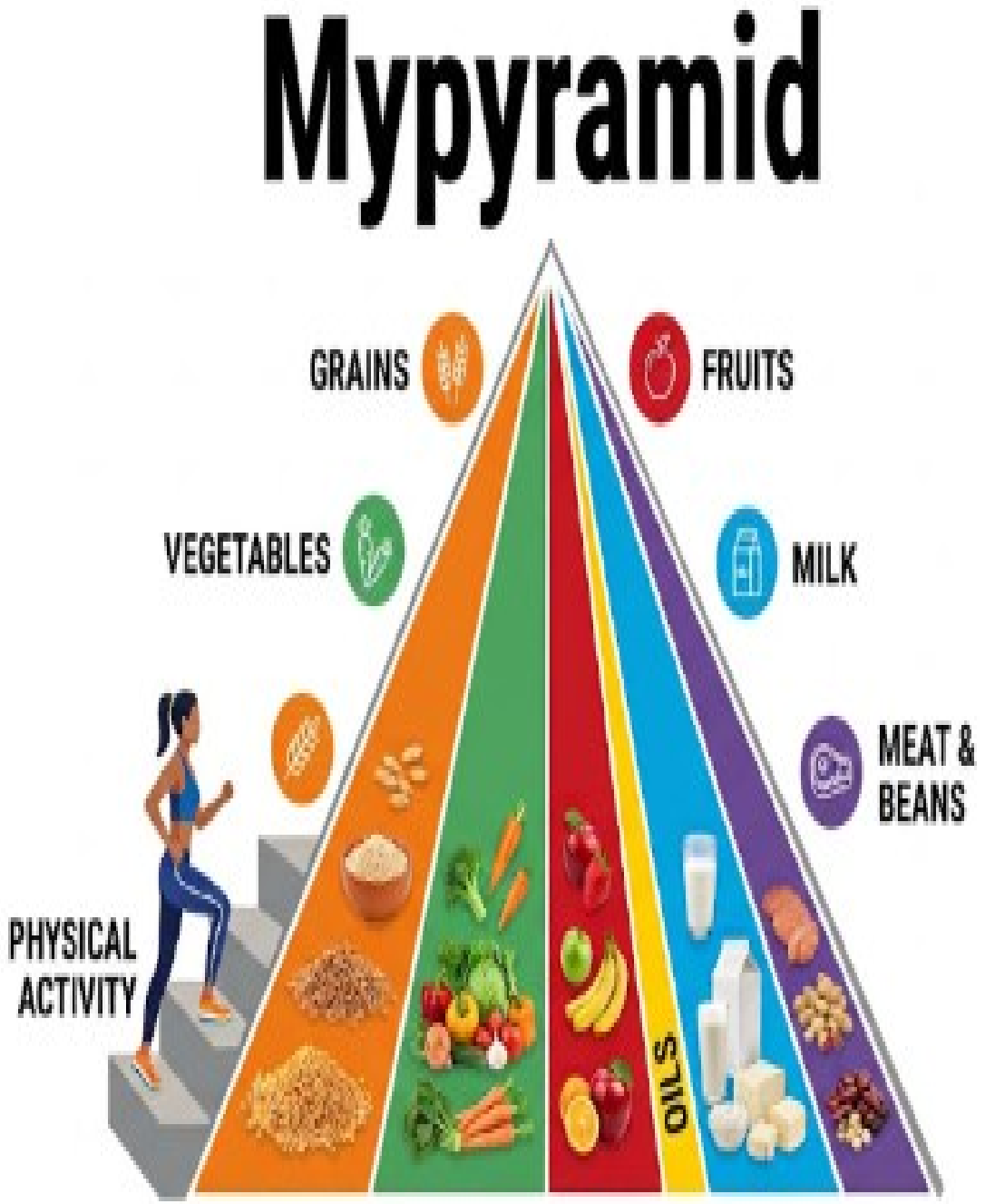
✓Then **fruits 2 to 4 servings/day**

✓Then **milk,** yogurt and cheese **2 to 3 servings/day**

✓Followed by **meat**, poultry, fish, dry beans, eggs, and nuts **2 to 3servings/day**

✓Finally, **fats, oils and sweets** in the small **apex** (to be used sparingly).

➢Food guide pyramid was **updated in 2005 to "MyPyramid", which contain colorful vertical wedges** that replace the hierarchical levels of the food guide pyramid.



➢Varying **width of the wedge** of food groups suggest that person should choose more from bigger wedges and vice versa however these bands do **not suggest exact proportion** they are just a general guide for proportion of food one should consumed. For example, an unmarked white tip represented items such as candy, alcohol, that should be taken sparingly

➢**Stairs** were added up the left side of the pyramid with an image of a climber to **represent a push for exercise**.

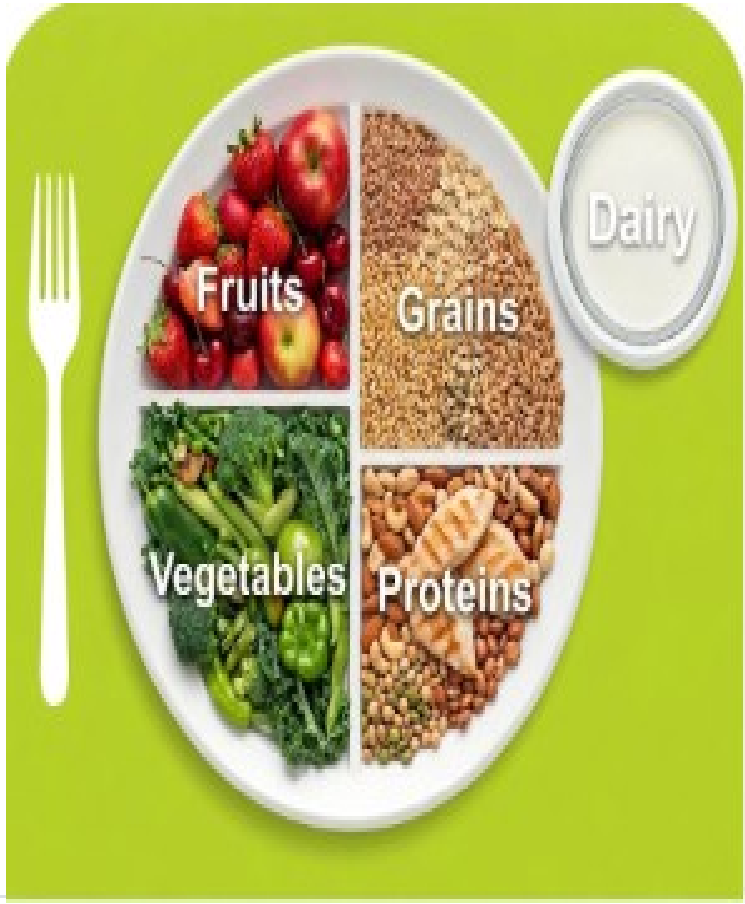
•And then **MyPyramid** diagram was replaced in 2011 by **"MyPlate".**

Which is the current nutrition guide published by the USDA consisting of a diagram of a plate and glass divided into **five food groups** to build a healthy plate:

•Fill 1/2 of plate with fruits and vegetables.

•The grains portion is slightly larger than the portion of protein foods.

•Having a serving of dairy on the side.



1 Balance calories.

Determine your calorie needs as a first step to melp in managing your weight. Go to www.ChooseMyPlate.gov to find your calorie level. Bsing physically active also helps you balance calories.

2 Enjoy food, but eat less.



Savor food, don't rush. Pour food as you eat it. Eating too fast or when your attention is elsewhere may lead to eating too many calories. Play attention to hunger and fullness cues before, eating, and after meats. Use them to recognize when to eat and when you've had enough.

3 Avoid oversized portions.

Use smaller plates. Bowl, and glass. Portion out foods before you eat. When eating out, choose a smaller size option, share a dish, or take home part of your meal.

4 Foods to eat more often.

Priot more vegetables, fruits, whole grains, and fat-free or 1 % milk and dairy products. These haves have the nutrients you need for health—including potassium, calcium, Vitamin D and fiber. Make them the basis for meals and snacks.



5 Make half your plate fruits and vegetables.

Choose diverse, deep-or dark-green vegetables like tomatoes, sweet potatoes, and drancell, along with other vegetables for your meals. Add fruit to meals as part of main or side dishen or as dessert.

6 Switch to fat-free or low-fat (1%) milk.



Same calcium, fewer calories and less fat.

7 Make half your grains whole grains.

Replace refined with whole grains (wheat, product a refined product—such as eating whole-wheat bread instead of white bread or brown rice instead of white rice.

8 Foods to eat less often.

Cut back on solid fats, solid fats, added sugars, and salt. They include cakes, cookies, ice cream, candies, sweetened drinks, pizza, and fatty meats like ribs, sausages, bacon, and hot dogs. Use these foods as occasional treats, not everyday foods.

9 Compare sodium in foods.



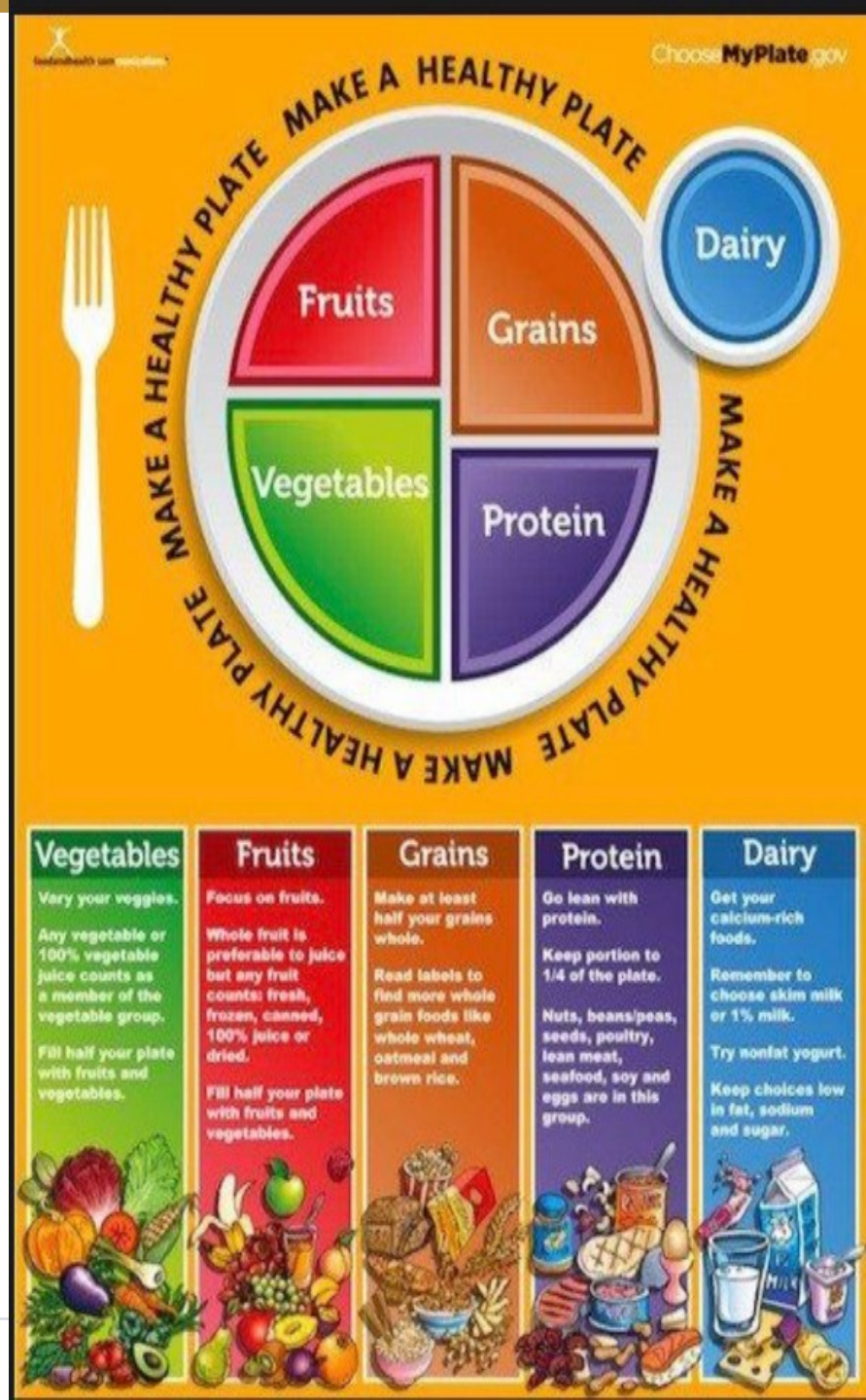
Check Nutrition Facts label to choose lower sodium versions of foods like soup, bread, and frozen noods. Select canned foods labeled "low sodium,, "reduced sodium," or "no salt added."

10 Drink water instead of sugary drinks

Cut calories by drinking water or unsweetened beverages. Soda, energy drinks, and sports drinks are a major source of added sugar, and calories, in American diets.

Additional recommendations of MyPlate include:

1. Balance energy
2. Enjoy food but eat less
3. Avoid oversized portions,
4. Make at least half (50%) of grains, whole grain
5. Choose a lean protein source.
6. Switch to fat-free or low-fat milk,
7. Choose foods with less sodium, and
8. Drink water instead of sugary drinks.

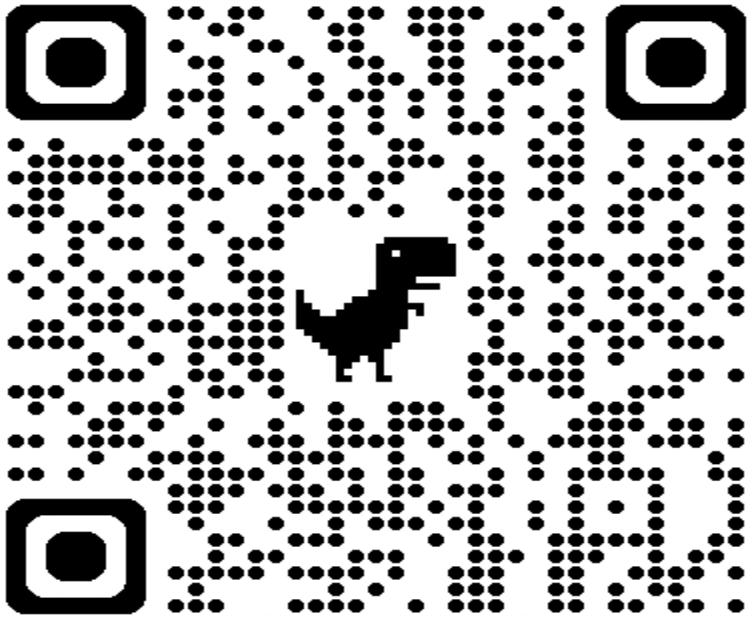


9.**Foods to eat more often** vegetables, fruits, whole grains, and fat-free or 1% milk and dairy products

10.**Foods to eat less often** foods high in solid fats, added sugars, and salt. They include cakes, cookies, ice cream, candies, sweetened drinks, pizza, and fatty meats like ribs, and sausages.

Summery:

1. Clinical nutrition is a medical specialty dealing with the relationship between disease & nutrition
2. Classification of nutrients according to chemical composition, quantity need, energy yielding, and essentiality.
3. Energy density measure of the energy a food provides relative to the amount of food vs Nutrient density refers to the amount of nutrients provided by a food relative to the amount of energy it provides.
4. Recommended Dietary Allowances is the daily intake of a nutrients that meets the needs of almost all (i.e., 97-98%) healthy individuals of a specific age and gender.
5. MyPlate is the current nutrition guide consisting of a diagram of glass of diary and a plate divided to fruits, vegetables ,grains, and protein groups of food



References

1. Nutrition and Diet Therapy 10th edition
2. Williams' Basic Nutrition and Diet Therapy 15 edition
3. Williams' Essentials of Nutrition and Diet Therapy 11th edition



QUESTIONS & DISCUSSION



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LECTURE COMPLETE